

Jojoy John

Postdoctoral Researcher | Microbiome, Microbial Ecology & Bioinformatics

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I am a microbiome researcher with expertise in genomics, metagenomics, metatranscriptomics, metabolic functional analysis, and statistical modeling of microbial communities. Currently working on interdisciplinary projects involving the microbiome recovery of estuarine & soil microbiomes, estuarine viromes, and screening genomes towards engineering of synthetic biofilms. Experienced in microbial ecology and OMICs data integration with strong computational and analytical skills. More details about my projects please see <https://jojoyjohn28.github.io/projects/>

Education

- **Postdoctoral Researcher**, Clemson University, USA — *Microbial Functional Redundancy* (2023–Present)
- **Ph.D. in Microbiology**, Sathyabama Institute of Science and Technology (SIST), India — *Genomic studies of halophilic bacteria* (2021)
- **Career Break (Maternity Leave)** — Completed an **online Master's in data science** (2021–2023)
- **M.Sc. & B.Sc. in Microbiology**, Mahatma Gandhi University, India (2014, 2012)

Research Experience

Postdoctoral Researcher

Clemson University, USA | 2023–Present

- Led **genome-resolved metagenomic and metatranscriptomic analyses** of estuarine microbiomes across seasonal and spatial gradients.
- Integrated **MAG-level metabolic potential with transcriptomic activity** to quantify functional redundancy and niche differentiation.
- Developed **reproducible HPC pipelines** for large-scale multi-omics processing and statistical analysis.
- Conducted **short-, long-read, and hybrid assembly of 200+ genomes** and screened them for synthetic community design.
- Performed **viromics analyses**, including viral contig identification, annotation, AMG detection, and virus–host interaction inference.
- Collaborated with interdisciplinary teams across microbiology, ecology, and computational biology.

Project Assistant II (Bioinformatics)

National Institute of Oceanography (NIO), Goa, India | 2019–2021

- Contributed to an **Indian government–funded project** assessing environmental impacts of deep-sea mining.
- Performed **16S/18S amplicon sequencing and shotgun metagenomics** of deep-sea and hypoxic environments (~600 m; Arabian Sea).

- Participated in **35-day oceanographic research cruises**, including field sampling and onboard DNA/protein preparation.
- Provided **Linux server maintenance and bioinformatics support** and assisted with user training.

Doctoral Research Fellow

Sathyabama Institute of Science and Technology (SIST), India | 2016–2019

- Conducted **whole-genome sequencing, comparative genomics, and core/pan-genome analyses** of moderate halophilic bacteria.
- Performed **16S rRNA gene sequencing (including Oxford Nanopore)** and phylogenetic analyses.
- Investigated **biodegradation and dye decolorization**, supported by **FTIR-based functional characterization**.
- Cultivated bacterial strains and performed **DNA, RNA, and protein extractions**.

Technical Expertise

1. **Genomics & Metagenomics:** Short-, long-read, and hybrid assembly; MAG reconstruction; co-assembly; ML-based binning
2. **Metatranscriptomics:** Read mapping, DNA/RNA integration, differential expression
3. **Viromics:** Viral contig identification, annotation, AMG detection, virus–host interaction, network analysis
4. **Metabolism & Modeling:** Co-occurrence/co-metabolism networks; linear & mixed models; multivariate ecology (Bray–Curtis, PERMANOVA, RDA/db-RDA, PCA/PCoA)
5. **Computing:** Python, R, Linux, Bash; HPC/cloud computing; Snakemake; Git

Publications

Accepted / In Press

1. **John, J.**, Ortiz, M., Ramond, P., & Campbell, B. J. (2026). Functional redundancy and metabolic flexibility of microbial communities in two Mid-Atlantic bays. *ISME Communications*. **Accepted**, awaiting proof.
2. **John, J.†**, Manigundan, K.†, Radhakrishnan, M., Kumar, A., Abirami, B., & Parli, V. B. (2026). Comparative genomic and phenotypic analysis of *Streptomyces rochei* SOSIST-3 isolated from the Southern Ocean. *Molecular Biology Reports*. **Accepted**; awaiting final proof. †Co–first authors
3. Ramadoss, D., **John, J.**, Badesab, F., Kocherla, M., Vasanth, A., & Mohandass, C. (2026). Baseline evaluation of the toxicity potential of steel slag using the green mussel *Perna viridis* for sustainable marine reuse applications. *Environmental Geochemistry and Health*. **Accepted**, awaiting proof.

Published (Peer-Reviewed Journals)

4. Ramadoss, D., Ammanabrolu, B. S., Acharya, A., **John, J.**, & Ingole, B. (2025). Deep-sea life associated with sediments and polymetallic nodules from the Central Indian Ocean Basin: Insights from 18S metabarcoding. *Deep Sea Research Part II: Topical Studies in Oceanography*, 105487.

5. Ramadoss, D., Biju, A., Rathore, C., Saha, M., Kolandhasamy, P., Palogi, C., **John, J.**, & Behera, A. K. (2025). The first report on emerged microplastics in deep-sea sediment: Insights from the Central Indian Ocean Basin. *Marine Pollution Bulletin*, 211, 117435.
6. **John, J.**, Dineshran, R., Hemalatha, K. R., Dhassiah, M. P., Gopal, D., & Kumar, A. (2020). Biodecolorization of synthetic dyes by a halophilic bacterium *Salinivibrio* sp. *Frontiers in Microbiology*, 11, 3281.
7. **John, J.**, Siva, V., Kumari, R., Arya, A., & Kumar, A. (2020). Unveiling cultivable and uncultivable halophilic bacteria inhabiting Marakkanam saltpan, India and their potential for biotechnological applications. *Geomicrobiology Journal*, 37(10), 1–11.
8. **John, J.**, Siva, V., Richa, K., Arya, A., & Kumar, A. (2019). Life in high salt concentrations with changing environmental conditions: Insights from genomic and phenotypic analysis of *Salinivibrio* sp. *Microorganisms*, 7(11), 577.
9. **John, J.**, Siva, V. S., & Kumar, A. (2020). Physiological tolerance of the early life history stages of freshwater prawn (*Macrobrachium rosenbergii*) to environmental stress. *Indian Journal of Geo-Marine Sciences*, 49(3), 382–389.

Conference Proceedings

10. **John, J.**, Siva, V., & Kumar, A. (2020). Isolation and characterization of saltpan bacteria for developing bacterial consortia for complete nitrogen removal from aquaculture ponds. In *An Insight on Environmental Toxicology* (ISBN: 978-93-5396-157-2), pp. 377–388.

Manuscripts Under Review / In Preparation

- 5 manuscripts under review
- 2 manuscripts in preparation for submission

Abstract and Conferences (selected since 2024)

1. Alisha M. Paul, **Jojoy John**, Diptee Chaulagain, David Karig, Barbara J. Campbell. 2025 Functional redundancy of marine synthetic biofilm communities under different environmental stresses. 2025, ASM biofilm, Oregon, USA
2. **Jojoy John**, Mir Alvee Ahmed, Barbara J. Campbell. 2025. Metabolic flexibility, secondary metabolites and seasonal dynamics in particle-associated and free-living microorganisms in the Chesapeake and Delaware Bays, ASM, Los Angeles, USA
3. Barbara J. Campbell, Kasey Kiser, Sam Stuckert and **Jojoy John**. 2025. Bacteriophages and Auxiliary Metabolic Gene interactions in microbial adaptations in the Chesapeake and Delaware Bays
4. **Jojoy John**, Maximiliano Ortiz, and Barbara J Campbell 2025. Functional Redundancy and metabolic flexibility of microbial communities in two mid Atlantic Bays. ASLO Aquatic Sciences Meeting, Charlotte, USA
5. **Jojoy John**, Maximiliano Ortiz, Pierre Ramond and Barbara J. Campbell (2024) Microbial functional redundancy in response to substrate and energy utilization in estuarine ecosystems 19th International Symposium on Microbial Ecology (ISME19). · Cape Town, South Africa.
6. **Jojoy John**, Maximiliano Ortiz, Barbara J. Campbell 2024. Does Microbiome Insure? Clemson University 3rd postdoctoral symposium; Clemson, South Carolina, USA

Mentoring & Student Supervision

Mentored PhD students, MSc students, and research interns in genome-resolved metagenomics, metatranscriptomics, viromics, and metabolic/statistical modeling, emphasizing reproducible workflows and independent analysis.

- Supervised or **co-mentored 5+ graduate students** (Clemson University, McGill University, SIST, India).
- Mentored **undergraduate and summer research interns** in bioinformatics and viromics.
- Co-author on student-led manuscripts.
- Provided ongoing **lab-wide bioinformatics support** and managed shared HPC/cloud infrastructure (Clemson Palmetto2).

Teaching Experience

- Co-Instructor, MICR 8130: Practical Bioinformatics for Microbiologists — Clemson University (2026)
- Instructor, Readings in Ecology — Clemson University (2024–2025)
- Co-Instructor, Hands-on Training in Metagenome Analysis — Clemson University (2023–2024)
- Laboratory Assistant, General Microbiology — SIST, India (2018–2019)
- Laboratory Assistant, Microbiology & Biotechnology — Mar Athanasius College, India (2013–2014)

Honors & Awards

- Best Poster Award, ICWEE (2018)
- Best Oral Presentation, OMICS-AFM Symposium (2018)
- 6th Rank, M.Sc. Microbiology — MG University (2014)
- 10th Rank, B.Sc. Microbiology — MG University (2012)

Service & Outreach

- Invited talks on microbial ecology and genomics (India, USA)
- Peer reviewer for 19 manuscripts (Microbiology Spectrum, AEM, mSphere, Scientific Reports, BMC Microbiology etc.)

References

1. **Barbara J. Campbell, Ph.D.**
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